Complex Capital Markets

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Simple Market Hypothesis

Popular (even if implicit) view: **there is a simple model of asset returns.**

- Asset returns can be explained with a small number of factors (e.g., CAPM, FF5)

Why is this view popular?

- **Parsimony:** Statistical and explanatory appeal of keeping models simple.
- **"Physics Envy"** (Mirowski, 1989): The drive in economics to discover elegant, low-dimensional "laws" of finance that mimic the fundamental laws of physics.

Complex Market Hypothesis

Asset returns can potentially be explained with a **large number** of factors or trading signals.

Is this view popular?

- **Industry: yes!** U.S. Senate Committee (2024): Majority of institutional investors use (complex) ML models.
- **Academia: "factor zoo."** For example, see Harvey vs. Pedersen debate on the "factor zoo" at the 2022 AFA.
- **Academia: "Virtue of Complexity."** See Kelly, Malamud, and Zhou (2024); Didisheim, Ke, Kelly, and Malamud (2024).

Equilibrium in simple markets

Rational expectations in simple markets:

- Classic Rational Expectations assumption: agents possess complete knowledge of the data-generating process.
  - They could learn a small number of parameters with a sufficient amount of data.
- "Dark matter" critique: even when models are simple, agents often know more about the parameters than they could possibly learn from the available data (Chen, Dou, and Kogan, *JF*, 2024).

Equilibrium in complex markets

We need a new theoretical foundation for equilibrium in complex markets:

- It is not possible to perfectly learn the complex DGP (e.g., impossible to estimate a model when the number of parameters exceeds the number of observations).
- Model uncertainty is first order.

Our Goal: Develop a theoretical foundation for the Complex Market Hypothesis.
Analogy: Grossman and Stiglitz (1980) and the Efficient Market Hypothesis.

The Model

- À la Grossman and Stiglitz (1980): two periods, one risky asset, CARA agents.
- $t \in \{0, 1\}$: $t = 0$ trade happens, $t = 1$ risky asset pays off d_1
- Complex Market Hypothesis

$$d_1 = b' s_1 + \beta' S_1 + \varepsilon_1,$$

  b is $n \times 1$, n is fixed. β is $N \times 1$, N is large.
- Model uncertainty: b, β and the respective covariance matrices of signals ψ $(n \times n)$ and Ψ $(N \times N)$ are not known.

The Model (cont.)

- Agents could learn about these parameters from the historical data $\mathcal{F}_0 = \sigma(d_\tau, s_\tau, S_\tau)$, $\tau = 1 - T, \ldots, 0$.
- DGP for fundamentals

$$d_\tau = b' s_\tau + \beta' S_\tau + \varepsilon_\tau$$

$\varepsilon_\tau \sim \mathcal{N}\left(0, \sigma_\varepsilon^2\right)$, $\quad s_\tau \sim \mathcal{N}(0, \psi)$, $\quad S_\tau \sim \mathcal{N}(0, \Psi)$

the sequences are i.i.d. over time, and s_τ is independent of S_τ for each τ.

b, β, ψ, and Ψ are fixed parameters. However, they are uncertain to agents.

The Model (cont.)

- Common prior:

$$b \sim \mathcal{N}\left(0, \sigma_b^2 I_n\right), \quad \beta \sim \mathcal{N}\left(0, \frac{\sigma_\beta^2}{N} I_N\right),$$

  b and β are independent
- Inverse-Wishart prior on covariance matrices ψ and Ψ, with $n+4$ and $N+4$ degrees of freedom and scale matrices $v_0 I_n$ and $V_0 I_N$, for some $v_0, V_0 \geq 0$. Here I_m is the $m \times m$ identity matrix.
- Given a Gaussian prior on β, the posterior is also Gaussian: $\beta \mid \mathcal{F}_0 \sim \mathcal{N}\left(\hat{\beta}_t, \hat{\Sigma}_t\right)$.
- The posterior on Ψ is also inverse-Wishart.
- Similarly for b and ψ.

The Model (cont.)

$$d_1 = b' s_1 + \beta' S_1 + \varepsilon_1,$$

Agents: unit continuum of CARA with risk aversion γ.

- **Uninformed** (fraction $1 - \lambda$): only know public information $\mathcal{F}_0$. *Face both model uncertainty and fundamental uncertainty.*
- **Informed** (fraction λ): know $\mathcal{F}_0$ as well as s_1 and S_1 (acquired at cost $\mathcal{C}$). *Only face model uncertainty.*
- Both uninformed and informed have the same prior.
- **Noise** traders submit $\sigma_\eta \eta_1$, where $\eta_1 \sim \mathcal{N}(0, 1)$.

We characterize the limiting REE as $N, T \to \infty$, $N/T \to c$.

Preview of Main Results

Key mechanism: interaction between fundamental uncertainty and model uncertainty in high dimensions.

This results in:

1. **Persistent uncertainty wedge:** Uninformed agents underestimate the uncertainty about fundamentals that informed agents face $\implies$ overestimate how aggressively informed trade.
2. **Tests of rational expectations fail for uninformed agents.** Negative Coibion–Gorodnichenko coefficient: forecast revision is negatively related to forecast error (even though uninformed agents are, by definition, rational).
3. **Priors matter.** Can choose priors (from a parametric family) to maximize objective expected utility. Optimal priors are different for informed vs. uninformed. No no-trade theorem with optimal priors.

Bayesian Learning: Simple Market ($\beta = 0$)

$d_\tau = b' s_\tau + \varepsilon_\tau$. Define shrinkage: $z_b = \sigma_\varepsilon^2 / \sigma_b^2$, signal matrix $s = (s_1, \ldots, s_T)'$.

- **Sample covariance matrix**: $\hat{\psi} = \frac{1}{T} s' s$.
- **Posterior mean**:

$$\hat{b} \equiv E_T[b] = \underbrace{\left(\frac{z_b}{T} I_n + \frac{1}{T} s' s \right)^{-1} \left(\frac{1}{T} s' \mathrm{d} \right)}_{\approx b_{\mathrm{OLS}}}$$

- **Posterior variance**:

$$\hat{\Sigma}_{bb} \equiv \mathrm{Var}_T[b] = \frac{\sigma_\varepsilon^2}{T} \left(\frac{z_b}{T} I_n + \frac{1}{T} s' s \right)^{-1}$$

- As $T \to \infty$: $\hat{\psi} \to \psi$, $\hat{b} \to b$, and $\hat{\Sigma}_{bb} \to 0$. Agents learn all parameters perfectly.

Bottom line: with fixed n, all parameters are learnable. No residual model uncertainty.

Bayesian Learning: Pure Complex Market ($b = 0$)

$d_\tau = \beta' S_\tau + \varepsilon_\tau$. Define shrinkage: $z_\beta = \sigma_\varepsilon^2 / \sigma_\beta^2$, signal matrix $S = (S_1, \ldots, S_T)'$.

- **Sample covariance matrix**: $\hat{\Psi} = \frac{1}{T} S'S$, $c = N/T$.
- **Posterior mean**:

$$\hat{\beta} \equiv E_T[\beta] = \underbrace{\left(z_\beta c\, I_N + \frac{1}{T} S'S \right)^{-1} \left(\frac{1}{T} S' \mathrm{d} \right)}_{\text{ridge regression with penalty } z_\beta c}$$

- **Posterior variance**:

$$\hat{\Sigma}_{\beta\beta} \equiv \mathrm{Var}_T[\beta] = \frac{\sigma_\varepsilon^2}{T} \left(z_\beta c\, I_N + \frac{1}{T} S'S \right)^{-1}$$

- As $T \to \infty$ with $N/T \to c$: $\hat{\Psi} \not\to \Psi$, $\hat{\beta} \not\to \beta$, and $\hat{\Sigma}_{\beta\beta} \not\to 0$.

Bottom line: persistent model uncertainty.

But how to calculate limits?

Random Matrix Theory

Use **Random Matrix Theory** to calculate the complex limits ($N, T \to \infty$, $N/T \to c$).

No miracles: we cannot consistently estimate

- the covariance matrix Ψ,
- the posterior covariance matrix $\hat{\Sigma}_{\beta\beta}$.

However, we often do not need to know each element of these matrices. Instead, the equilibrium relies on the macroscopic behavior of these matrices. RMT gives us the **limiting spectral distributions**, which allow us to compute the needed macroscopic summaries of these large matrices.

RMT: Concentration of Quadratic Forms [1/3]

Definition

Given two functions $X = (X_T),\ Y = (Y_T)$ of $(S_\tau, d_\tau)_{1 \le \tau \le T}$, we write $X \approx Y$ if

$$\operatorname*{plim}_{N,T \to \infty,\ N/T \to c} (X_T - Y_T) = 0$$

for any $c > 0$.

RMT: Concentration of Quadratic Forms [2/3]

Lemma (Concentration of Quadratic Forms)

Let A be a uniformly bounded (random) matrix that is independent of S_1, and let

$$Y = N^{-1} S_1' A S_1.$$

Then,

$$Y \approx N^{-1} \operatorname{tr}(A\Psi).$$

Illustration: $\Psi = I_N$, $A = I_N$ (i.e., $S_{i,1}$ are i.i.d. with $E[S_{i,1}^2] = 1$). Then,

$$Y = N^{-1} S_1' S_1 = N^{-1} \sum_{i=1}^{N} S_{i,1}^2 \xrightarrow{p} E[S_{i,1}^2] = 1 = N^{-1} \operatorname{tr}(I_N).$$

This is just the **Law of Large Numbers in** N.

RMT: Concentration of Quadratic Forms [3/3]

Key idea: in high dimensions, complex random objects concentrate around simpler deterministic limits.

$$\underbrace{N^{-1} S_1' A\, S_1}_{\text{depends on } S_1} \approx \underbrace{N^{-1} \operatorname{tr}(A\Psi)}_{\text{does not depend on } S_1}$$

- If A is random, $N^{-1} \operatorname{tr}(A\Psi)$ can be replaced with a **deterministic equivalent** that can be computed using only the eigenvalue distribution of A.

General RMT insight: scalars involving large random matrices have deterministic limits.

- At scale, large random matrices self-average: micro-level noise cancels out to yield deterministic macroscopic scalars. RMT allows us to calculate these limits.

RMT: Application to Informed Variance

Recall $d_1 = b' s_1 + \beta' S_1 + \varepsilon_1$.

Informed agent's conditional variance:

$$\sigma_I^2 \equiv \mathrm{Var}[d_1 \mid s_1, S_1] = \sigma_\varepsilon^2 + s_1' \hat{\Sigma}_{bb}\, s_1 + \underbrace{S_1' \hat{\Sigma}_{\beta\beta}\, S_1}_{\text{depends on } S_1}$$

By the concentration lemma ($\hat{\Sigma}_{bb} \to 0$ as $T \to \infty$):

$$\sigma_I^2 \approx \sigma_\varepsilon^2 + \underbrace{\mathrm{tr}(\hat{\Sigma}_{\beta\beta}\, \Psi)}_{\text{independent of } S_1}$$

The complex part concentrates and the simple part vanishes: σ_I^2 becomes **independent of the signal realization**.

Persistent Uncertainty Wedge

Recall $z_\beta = \sigma_\varepsilon^2/\sigma_\beta^2$ (shrinkage). Define:

$$\sigma_I^2 = \mathrm{Var}[d_1 \mid s_1, S_1], \qquad \hat{\sigma}_I^2 = E[\sigma_I^2 \mid \mathcal{F}_0].$$

Proposition (Uncertainty Wedge)

In the limit $T, N \to \infty$ with $N/T \to c > 0$ and n fixed,

$$\sigma_I^2 \;\approx\; \sigma_I^2(z_\beta, c), \qquad \hat{\sigma}_I^2 \;\approx\; \hat{\sigma}_I^2(z_\beta, c),$$

for some explicit functions $\sigma_I^2(z_\beta, c)$ and $\hat{\sigma}_I^2(z_\beta, c)$. Furthermore,

$$\hat{\sigma}_I^2(z_\beta, c) \;<\; \sigma_I^2(z_\beta, c).$$

Uninformed agents underestimate the risk that informed agents face.

Persistent Uncertainty Wedge: Implications

Why does it matter that $\hat{\sigma}_I^2(z_\beta, c) < \sigma_I^2(z_\beta, c)$?

- Uninformed filter information from prices by estimating how aggressively informed trade.
- Lower perceived risk $\implies$ they believe informed trade **more aggressively** than they actually do.
- This distorts uninformed inference from prices, with consequences for equilibrium.

Persistent Uncertainty Wedge: Intuition

Uninformed agents face **two sources of uncertainty** when predicting $\mathrm{Var}[d_1 \mid s_1, S_1]$:

1. **Model uncertainty:** they don't know β.
2. **Fundamental uncertainty:** they don't know S_1.

These two sources of uncertainty **offset each other**:

- If signals vary less $\implies$ less fundamental uncertainty, but signal variation is what allows agents to estimate β. So less fundamental uncertainty means **more model uncertainty**, and vice versa.
- Formally: $\hat{\Psi}$ and $\hat{\Sigma}_{\beta\beta}$ are negatively related, because $\hat{\Sigma}_{\beta\beta} = \frac{\sigma_\varepsilon^2}{T}(z_\beta c\, I_N + \hat{\Psi})^{-1}$.

This negative correlation biases $\hat{\sigma}_I^2$ downward.

Equilibrium: Informed Traders

Informed agents observe s_1 and S_1. Given their posterior, b and β are normally distributed $\implies$ CARA-Normal $\implies$ standard mean-variance optimization:

$$x_I(P) \approx \frac{b' s_1 + \hat{\beta}' S_1 - P}{\gamma \sigma_I^2}$$

Note: $\hat{b} \to b$ (learned perfectly), $\sigma_I^2 \approx$ const by concentration.

Market clearing:

$$\lambda x_I(P) + (1-\lambda) x_U(P) = \sigma_\eta \eta_1$$

What uninformed can learn from the price:

$$\iota = \underbrace{b' s_1 + \hat{\beta}' S_1}_{\text{information}} - \underbrace{\psi \eta_1}_{\text{noise}}, \qquad \psi = \frac{\sigma_I^2 \gamma \sigma_\eta}{\lambda}$$

Equilibrium: Uninformed Traders

Challenge: for finite T, N, the payoff $d_1 = b's_1 + \beta' S_1 + \varepsilon_1$ is **not normal** conditional on $(\iota, \mathcal{F}_0)$.

- d involves a product of two normals (β and S_1).
- So $d_1 \mid \iota, \mathcal{F}_0$ has a non-Gaussian distribution $\implies$ CARA-Normal optimization does not apply.

Lemma

In the limit as $T, N \to \infty$, $N/T \to c$, the conditional distribution $d_1 \mid \iota, \mathcal{F}_0$ is normal.

Uninformed demand:

$$x_U(P) \approx \frac{E[d_1 \mid \iota, \mathcal{F}_0] - P}{\gamma \operatorname{Var}[d_1 \mid \iota, \mathcal{F}_0]}$$

Equilibrium

Theorem

There exists a unique equilibrium. The share of informed agents λ satisfies:

- $\lambda = 1$ *if* $\mathcal{C}$ *is low enough,*
- $\lambda = 0$ *if* $\mathcal{C}$ *is high enough,*
- $\lambda \in (0, 1)$ *otherwise.*

Limiting equilibrium quantities can be expressed in closed form via the Stieltjes transform *of the eigenvalue distribution of* Ψ *and other model parameters.*

Tests of Rational Expectations

Coibion–Gorodnichenko regression

$$\underbrace{d_1 - E[d_1 \mid \mathcal{F}_0, P]}_{\text{forecast error}} = \alpha + \beta_{CG} \underbrace{E[d_1 \mid \mathcal{F}_0, P] - E[d_1 \mid \mathcal{F}_0]}_{\text{forecast revision}} + u$$

Note that $E[d_1 \mid \mathcal{F}_0] = 0$. The population coefficient β_{CG} is given by:

$$\beta_{CG} = \frac{E^*\Big[\big(d_1 - E[d_1 \mid \mathcal{F}_0, P]\big)\, E[d_1 \mid \mathcal{F}_0, P]\Big]}{\mathrm{Var}^*\Big(E[d_1 \mid \mathcal{F}_0, P]\Big)}$$

where E^* denotes the expectation under the *objective* measure in which β and Ψ are fixed (known) parameters.

Premise of the test: Under rational expectations, $\beta_{CG} = 0$.

CG Regression in Complex Markets

Uninformed agents are rational—Bayesian learners with common priors. Yet:

Proposition. In the limit, $\beta_{CG}^{U} < 0$.

Intuition.

- The forecast has the form $E[d_1 \mid \mathcal{F}_0, P] = \kappa\, \iota$, where κ depends on estimated parameters.
- Under the objective measure, $E^*[d_1 \mid \mathcal{F}_0, P] = \kappa^*\, \iota$, with $\kappa^* < \kappa$.
  (Uninformed overestimate how aggressively informed trade $\Rightarrow$ overestimate price informativeness.)
- By construction, κ^* satisfies $E^*\big[\big(d_1 - \kappa^* \iota\big)\, \kappa^* \iota\big] = 0$.
- Replacing κ^* with $\kappa > \kappa^*$:

$$E^*\big[\big(d_1 - \kappa\, \iota\big)\, \kappa\, \iota\big] < 0 \quad \Longrightarrow \quad \beta_{CG}^{U} < 0.$$

Is It Good to Be Bayesian?

Agents maximize $\mathbb{E}[u \mid \mathcal{F}]$, where $\mathbb{E}$ is the expectation under the *epistemic* measure:

- They do not know β, Ψ and treat them as random according to their prior.

Objective expected utility: $\mathbb{E}^*[u]$, where $\mathbb{E}^*$ is the expectation under the *objective* probability measure:

- Parameters β, Ψ are fixed.

Because model uncertainty **persists**, the two do not coincide: $\mathbb{E}^*\big[\mathbb{E}[u \mid \mathcal{F}]\big] \neq \mathbb{E}^*[u]$.

Being Bayesian can hurt. There exist priors (i.e., shrinkage z) such that:

- Objective expected profit for uninformed agents can be **negative**.
- Objective expected profit for informed agents, **net of information costs**, can also be **negative**.

Do Priors Matter?

Simple markets (low-dimensional): being Bayesian does not hurt.

- As long as the prior is wide enough to include the true parameter, the posterior converges to the truth as $T \to \infty$.
- The wedge $\mathbb{E}^*\big[\mathbb{E}[u \mid \mathcal{F}]\big] - \mathbb{E}^*[u] \to 0$.
- Priors wash out $\implies$ the specific choice of prior is immaterial.

Complex markets (high-dimensional): priors matter.

- Model uncertainty persists even as $T \to \infty$ (with $N/T \to c$).
- The wedge does **not** vanish $\implies$ the choice of prior has lasting consequences.

Can we choose priors optimally?

Optimal Prior

Suppose agents are **hyper-rational**: they choose their prior optimally.

The prior parameter that matters in the limit is the shrinkage

$$z = \frac{\sigma_\varepsilon^2}{\sigma_\beta^2}.$$

Optimization problem: agents choose z to maximize objective expected utility.

1. For a given z, their demand maximizes *epistemic* expected utility $\mathbb{E}[u \mid \mathcal{F}]$.
2. Evaluate the resulting demand under the *objective* measure: compute $\mathbb{E}^*[u]$.
3. Choose z to maximize $\mathbb{E}^*[u]$.

The problem is well-defined: objective expected utility does not depend on Ψ itself—only on the eigenvalue distribution of Ψ, which can be consistently estimated.

Optimal Priors

Recall $z = \sigma_\varepsilon^2/\sigma_\beta^2$. Two polar cases:

- **Simple Market Hypothesis:** $\beta = 0$ as a prior $\iff \sigma_\beta^2 \to 0 \iff z \to \infty$.
- **Complex Market Hypothesis:** finite z.

Result:

- $z \to \infty$ (simple market prior) is **never the worst**, but **never the best**. The optimal prior is always finite: $z^* < \infty$. Some finite z are worse than $z \to \infty$.

Implications for trade:

- Optimal priors can be chosen *after* the signal is acquired
- Informed and uninformed choose **different** optimal priors $\implies$ possible to sustain trade **without noise**.

Conclusion

Complex Market Hypothesis: a large number of trading signals are relevant for returns.

1. **Persistent model uncertainty:** parameters cannot be fully learned $\implies$ epistemic and objective measures diverge.
2. **Uncertainty wedge:** uninformed agents underestimate the risk informed agents face $\implies$ overestimate price informativeness.
3. **Tests of rationality fail:** negative Coibion–Gorodnichenko coefficient for rational Bayesian agents.
4. **Being Bayesian can hurt:** objective expected profits can be negative for both informed and uninformed.
5. **Priors matter and can be optimized:** informed and uninformed choose different optimal priors $\implies$ trade without noise.